Preview: Section 2.1 Datasets (text above, table below)

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [17] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSM-Toxin [18], ToxinPred2 [19], and ATSE [20], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [21] using the query "NOT KW-0800 AND NOT KW-0020 AND reviewed:true", and sequences containing non-standard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [22] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following [2], 85% of the data was used as the training set and the remaining 15% as the independent Test1 set, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic Test1 peptides. An additional independent Test2 set, containing 46 toxic and 536 non-toxic peptides, was used to further evaluate generalization ability. In addition, to assess the model on a strict homology-reduced external benchmark, we derived an external test set from the independent split of the ToxinPred 3.0 dataset [23] by removing sequences sharing at least 80% sequence identity with any training reference (CD-HIT-2D), which retained 977 peptides (471 toxic and 506 non-toxic). Table 1 summarizes the statistics of all datasets used in this study.

Table 1: Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
Test1 (benchmark)	320	806	1,126	Independent test
Test2	46	536	582	Independent test
ToxinPred 3.0 strict	471	506	977	Strict external test